tokentax: Measuring the Tokenizer Cost Penalty Across 203 Languages

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Abstract

The same sentence costs 1× tokens in English but roughly 9–11× in Shan, Santali, or Odia under current LLM tokenizers, because tokenizer vocabularies are trained on English-heavy corpora. Petrov et al. [2023] and Ahia et al. [2023] established this "token tax" for 2023-era tokenizers, but no maintained, versioned resource reports current numbers for the tokenizers actually in production use today. We present `tokentax`, a measurement suite, versioned results dataset, and library that computes the token premium, and its 95% confidence interval via `evalci` [Chandrahas, 2026], for 8 tokenizers (GPT-4o, GPT-4, GPT-2, Llama 3, Qwen2.5, DeepSeek-V3, Mistral, Gemma-2) across all 203 non-English FLORES-200 languages [NLLB Team et al., 2022]. The pipeline reproduces Petrov et al.'s published GPT-2/GPT-4 premium ratios within 1.1% relative error on five languages, a direct validation against prior work rather than only internal self-consistency. Across the seven currently-deployed tokenizers (excluding the 2019 GPT-2 vocabulary, which we retain only as a historical anchor), the mean premium is 2.74× and per-tokenizer means range from 2.13× (Gemma-2) to 3.16× (Mistral). Vocabulary size predicts the premium better than release year does (Spearman $\rho = -0.91$ vs. -0.76), and Mistral – a 2024 tokenizer with the smallest vocabulary in our set – carries a higher premium than either 2024 tokenizer with a 200k+ vocabulary, so the improvement over time is better read as a vocabulary-size effect than a release-date effect. A domain-robustness check for the 30 highest-premium languages finds the premium replicates closely on mixed-domain OPUS-100 text (median signed difference -1.7%) but runs systematically *lower* on a Bible-translation corpus (median -17%, and lower in all 112 comparisons), a register effect we report as a directional finding rather than as noise. 9 of those 30 languages have no modern, ungated parallel corpus on Hugging Face for a second domain at all – a scarcity that is itself part of the token-tax story. `tokentax` is available at `https://pypi.org/project/tokentax/` (source: `https://github.com/Shreyaskc/token-tax`), with the full results dataset and an interactive explorer at `https://huggingface.co/datasets/shreyaskc/tokentax-results-v1` and `https://huggingface.co/spaces/shreyaskc/tokentax-explorer`.

1 Introduction

Token-based API pricing and fixed context windows mean tokenizer inefficiency is a real economic and capability penalty, not a cosmetic one. If a language costs $k\times$ as many tokens as English for the same content, its speakers pay $k\times$ as much per query, retain $1/k$ as many effective few-shot examples in a fixed context window, and hit context truncation $k\times$ sooner. Petrov et al. [2023] first quantified this systematically, finding tokenization-length disparities of up to 15× across languages for tokenizers available in 2023; Ahia et al. [2023] independently showed that OpenAI's commercial API pricing, applied uniformly per token, overcharges speakers of many supported languages for equivalent utility. Rust et al. [2021] showed the same underlying disparity degrades downstream monolingual task performance, not just cost.

These are foundational results, but the tokenizers they measured are no longer the ones in production use. GPT-2's tokenizer (2019) and cl100k_base (2022, GPT-3.5/GPT-4) have both been superseded by larger, more multilingual-aware vocabularies – OpenAI's o200k_base (GPT-4o), and open-weight vocabularies from Meta, Alibaba, DeepSeek, Mistral, and Google. Whether the token tax has actually improved with these newer vocabularies, and by how much, is a question every multilingual-fairness paper since 2023 has had to re-derive ad hoc, because no maintained, versioned, statistically rigorous resource reports current numbers. This is precisely the gap a living benchmark should fill: the numbers change with every tokenizer release, which argues for a re-runnable measurement suite and a versioned dataset over a one-off paper.

We present `tokentax`: (1) a small Python library that computes the token premium, bytes/characters-per-token, and effective context-window shrinkage for any tokenizer against any parallel corpus, with confidence intervals via `evalci` [Chandrahas, 2026]; (2) a full sweep of 8 current tokenizers against all 203 non-English FLORES-200 languages, validated by reproducing Petrov et al.'s published numbers; (3) a domain-robustness check against a second corpus for the 30 most heavily taxed languages; and (4) a versioned results dataset and interactive explorer, so the resource stays current as new tokenizers ship.

2 Related Work

Petrov et al. [2023] is the direct precedent for this work: a systematic study of tokenization-length disparity across languages for GPT-2, GPT-3, GPT-4, and several open tokenizers of the time, using FLORES-200 as the parallel corpus – the same choice we make, for the same reason (Section 3). Ahia et al. [2023] took the complementary economic angle, translating token counts into actual API dollar costs and utility-per-dollar across 22 languages, and is the direct precedent for our cost-analysis framing (Section 5.5). Rust et al. [2021] showed that a designated monolingual tokenizer, not just pretraining data volume, materially affects downstream task performance – the capability-side consequence of the same disparity we measure on the cost side. NLLB Team et al. [2022] introduced FLORES-200, the parallel-sentence corpus this entire line of work, including ours, depends on.

`tokentax`'s contribution relative to this line of work is not a new finding about *why* the disparity exists – that question is settled – but a maintained instrument for *how much* it currently is, built to be re-run as tokenizers change, with every reported number carrying a confidence interval rather than a bare point estimate, and with an explicit domain-robustness check rather than a single-corpus measurement.

3 The `tokentax` Suite

Premium ratio. For a tokenizer T and an aligned sentence pair (s_{en}, s_ℓ) expressing the same content in English and language ℓ, the per-sentence premium ratio is $r_i = |T(s_\ell^{(i)})|/|T(s_{\text{en}}^{(i)})|$, where $|T(\cdot)|$ is token count. We compute this only on *aligned parallel* sentences, never independent monolingual corpora: only a shared-meaning pair makes "language ℓ costs $k\times$ as many tokens as English for the same content" a defensible claim, since monolingual corpora in different languages differ in register, sentence length, and content independent of tokenization efficiency.

Estimand. Every premium we report is the *mean of per-sentence ratios*, $\bar{r} = \frac{1}{n}\sum_i r_i$, which answers "what does a typical sentence cost?" and admits a straightforward per-sentence bootstrap. This is deliberately *not* the ratio of corpus totals, $\sum_i |T(s_\ell^{(i)})|/\sum_i |T(s_{\text{en}}^{(i)})|$, which answers a different question ("what does the whole corpus cost?") and weights long sentences more heavily. The two differ: by Jensen's inequality $\bar{r}$ is upward-biased relative to the ratio of totals, and across the ten (language, tokenizer) pairs of Section 4 we measure that bias at 0.9–1.7%. This matters when comparing our numbers to prior work, which generally reports the ratio of totals; see Section 4. Readers needing the other estimand can recompute it from the released per-sentence data.

Corpus. We use FLORES-200 devtest [NLLB Team et al., 2022]: 1,012 sentences, professionally translated in parallel into 200+ languages, covering 203 languages with an English pairing in our sweep. FLORES-200 is gated on Hugging Face (a free, instant license click, not a manual review).

Tokenizers. We study 8 tokenizers, loaded via `tiktoken` for OpenAI vocabularies and Hugging Face `transformers` [Wolf et al., 2020] for every open-weight one: OpenAI's o200k_base (GPT-4o) and cl100k_base (GPT-4), and the legacy GPT-2 vocabulary (kept only for the validation in Section 4); Meta's Llama 3; Alibaba's Qwen2.5; DeepSeek-V3; Mistral (v0.3); and Google's Gemma-2. Two deviations from the obvious choice, both discovered by actually running the pipeline rather than assumed in advance: `meta-llama/Meta-Llama-3-8B` requires Meta's manual license approval rather than an instant click-through, so our Llama 3 tokenizer resolves to the widely-used, ungated `NousResearch/Meta-Llama-3-8B` mirror of the same vocabulary (we have not independently verified byte-for-byte equivalence against the gated original; see Limitations); and Mistral's tokenizer turned out not to be gated at all once tested, only missing a `protobuf` runtime dependency. Anthropic's Claude is excluded: it has no downloadable tokenizer artifact, only a token-counting API endpoint, which would require every user reproducing this dataset to hold an API key. Gemma-2's SentencePiece vocabulary is also the closest available open proxy for Gemini's tokenizer, which likewise has no public artifact; we report it only under the Gemma-2 name and do not present it as a Gemini measurement.

GPT-2 is included as a *historical anchor* for the trend analysis (Section 5.3) and for the validation in Section 4; it is not a currently-deployed tokenizer. We therefore report aggregate statistics over the seven current tokenizers separately from the full set of eight wherever the distinction changes the number, and say which basis is in use each time.

Table 1 lists vocabulary sizes, which Section 5.3 uses to separate vocabulary-size effects from release-date effects.

Tokenizer	Family	Vocab. size	Mean premium
Mistral (v0.3)	Mistral	32,768	3.16×
GPT-2†	OpenAI	50,257	4.48×
GPT-4 (cl100k_base)	OpenAI	100,277	3.42×
Llama 3	Meta	128,256	3.04×
DeepSeek-V3	DeepSeek	128,815	2.52×
Qwen2.5	Alibaba	151,665	2.79×
GPT-4o (o200k_base)	OpenAI	200,019	2.15×
Gemma-2	Google	256,000	2.13×

Table 1: Tokenizers studied, ordered by vocabulary size, with mean premium over 203 non-English FLORES-200 languages. †GPT-2 is a 2019 historical anchor, not a currently-deployed tokenizer, and is excluded from "current tokenizer" aggregates.

Confidence intervals. Every reported premium ratio is a percentile bootstrap 95% confidence interval over the 1,012 per-sentence ratios, computed by `evalci.ci(method="bootstrap")` [Chandrahas, 2026] – not reimplemented, per this project's cross-artifact reuse policy.

4 Validation Against Prior Work

Rather than validate only against our own pipeline's internal consistency, we reproduce Petrov et al.'s own published per-language token totals over FLORES-200 devtest for GPT-2 and GPT-4's cl100k_base, for five languages spanning a range of scripts and premium magnitudes.[1] Table 2 shows agreement within 1.1% relative error for every (language, tokenizer) pair – well inside the 2% target we set before running it, and consistent with the small differences expected from FLORES-200 dataset revisions since 2023.

One caveat deserves emphasis, because it is easy to overstate what this table shows. To compare like with like, the reproduction column uses the *ratio of corpus totals*, matching how Petrov et al.'s released data is organized – not the mean-of-per-sentence-ratios estimand used for every other number in this paper (Section 3). What Table 2 establishes is therefore that our *tokenization and corpus handling* agree with prior work to within 1.1%; it does not independently validate the mean-of-ratios estimator, which by construction runs 0.9–1.7% higher on these same pairs. Both quantities are computable from the released per-sentence data, and we report which is which rather than blurring them into a single "validated" number.

Language	Tokenizer	Published	Reproduced*	Rel. error
Amharic	GPT-2	7.787	7.702	1.1%
Amharic	GPT-4	7.678	7.593	1.1%
Tamil	GPT-2	15.582	15.537	0.3%
Tamil	GPT-4	7.651	7.644	0.1%
Burmese	GPT-2	16.891	16.813	0.5%
Burmese	GPT-4	11.702	11.618	0.7%
Hindi	GPT-2	7.459	7.413	0.6%
Hindi	GPT-4	4.794	4.759	0.7%
Standard Arabic	GPT-2	4.400	4.358	1.0%
Standard Arabic	GPT-4	3.037	3.027	0.3%

Table 2: Reproduction of Petrov et al. [2023]'s published premium ratios. Maximum relative error 1.1% across 10 (language, tokenizer) pairs. *Computed as the ratio of corpus totals to match the reference data, not the mean-of-per-sentence-ratios estimand used elsewhere in this paper; the latter runs 0.9–1.7% higher on these pairs.

5 Results

5.1 Distribution

Across the seven current tokenizers × 203 languages (1,421 pairs), the mean premium is 2.74×, the median 2.08×, and the range spans 0.94× (Mandarin under DeepSeek-V3) to 15.33× (Shan under GPT-4). Including the GPT-2 historical anchor raises the mean to 2.96× and extends the maximum to 19.12×; we quote the seven-tokenizer figures throughout unless stated otherwise, since GPT-2 is not a tokenizer anyone deploys today.

The single sub-1× value is Mandarin Chinese under DeepSeek-V3 (0.94×, 95% CI [0.93, 0.95], excluding 1) – a genuine exception rather than noise, plausibly because a vocabulary trained with

[1] Reference values computed from `https://github.com/AleksandarPetrov/tokenization-fairness`, the paper's own released data – inferred from how that data is organized, not confirmed against their paper text; the close agreement below is evidence the inference is right.

substantial Chinese-language data can represent common multi-character sequences more compactly per token than the corresponding English byte-pair merges do. Notably, under current tokenizers the three lowest-premium languages are all Chinese variants (Simplified 1.30×, Cantonese 1.40×, Traditional 1.43×), displacing the Latin-script European languages that occupy those positions when GPT-2 is averaged in – a concrete instance of newer vocabularies redistributing the tax rather than uniformly lowering it.

5.2 Which languages are taxed the most

Figure 1 shows the 15 languages with the highest mean premium across the seven current tokenizers. All 11 languages above 6.5× are written in non-Latin scripts: Shan (Myanmar script), Santali (Ol Chiki), Odia, Dzongkha and Tibetan (Tibetan script), Tamasheq and Tamazight (Tifinagh), Lao, Burmese, Khmer, and Malayalam. At the opposite end, the 10 lowest-premium languages (1.30–1.51×) are Latin-script European languages plus Indonesian and the three Chinese variants.

We deliberately stop at describing this as a script-level pattern. It is tempting to explain it by how much text in each script appears in tokenizer pretraining corpora, and that is very likely the mechanism [Petrov et al., 2023, Rust et al., 2021]; but we measure tokenizer output, not training data, and the pretraining mixtures for six of our eight tokenizers are not public. Establishing the mechanism would require corpus-composition data we do not have, so we report the association and leave the causal claim to work that can measure the input side directly.

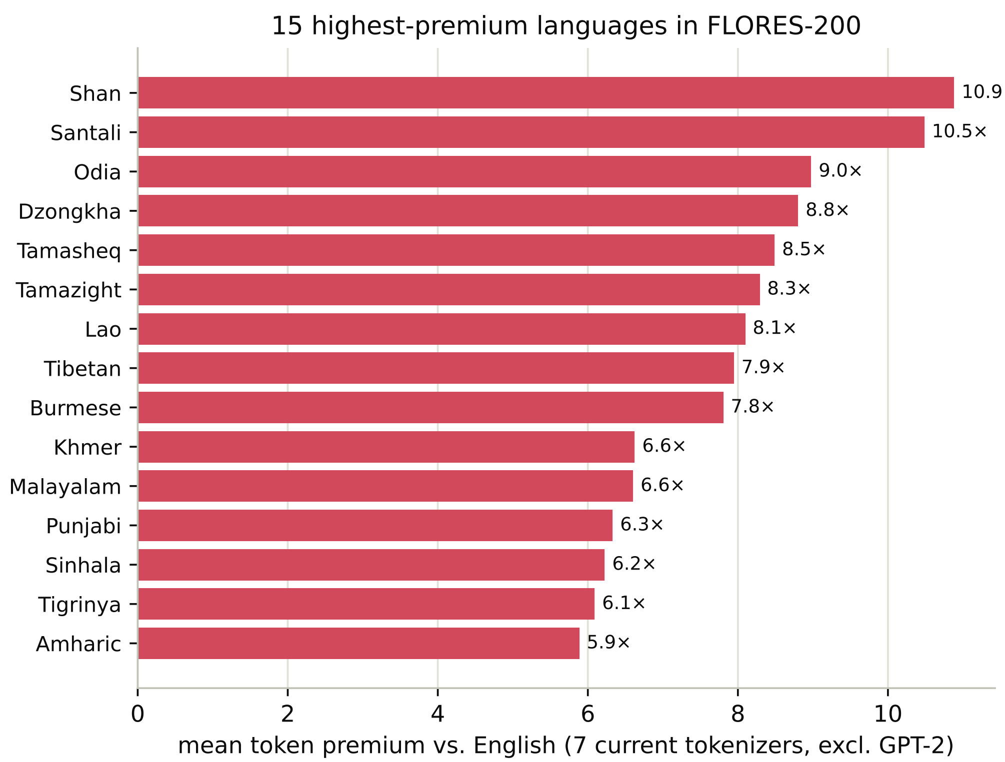

Figure 1: 15 languages with the highest mean token premium vs. English, averaged across the seven current tokenizers (GPT-2 excluded).

5.3 Vocabulary size, not release year

Figure 2 orders the 8 tokenizers by mean premium. All six 2024 tokenizers sit below GPT-4's 2022 cl100k_base (3.42×) and GPT-2's 2019 vocabulary (4.48×), and four of them (Gemma-2, GPT-4o, DeepSeek-V3, Qwen2.5) come in at ≤2.8×. That invites the summary "newer tokenizers are fairer." The summary is too coarse, and the vocabulary sizes in Table 1 show why.

Vocabulary size is the better predictor. Across the 8 tokenizers, mean premium correlates with vocabulary size at Spearman $\rho = -0.91$ ($p = 0.002$) but with release year at only $\rho = -0.76$ ($p = 0.027$); Figure 3 plots the former. The decisive case is Mistral: a 2024 tokenizer with the smallest vocabulary in our set (32,768) and a mean premium of 3.16× – barely better than the two-year-older GPT-4 vocabulary (3.42×), and roughly 50% worse than the two 2024 tokenizers with 200k+ vocabularies (2.13–2.15×). Being recent buys a language little if the vocabulary is small.

Two caveats keep this from being a clean causal claim. First, vocabulary size is itself correlated with how much multilingual data went into training the vocabulary, and we cannot separate the two from tokenizer artifacts alone. Second, both pre-2024 baselines (GPT-2, GPT-4) come from a single vendor, so "older" and "OpenAI" are partly confounded in the temporal comparison – though notably OpenAI's own o200k_base is among the lowest-premium vocabularies we measured, which cuts against reading the trend as a vendor effect.

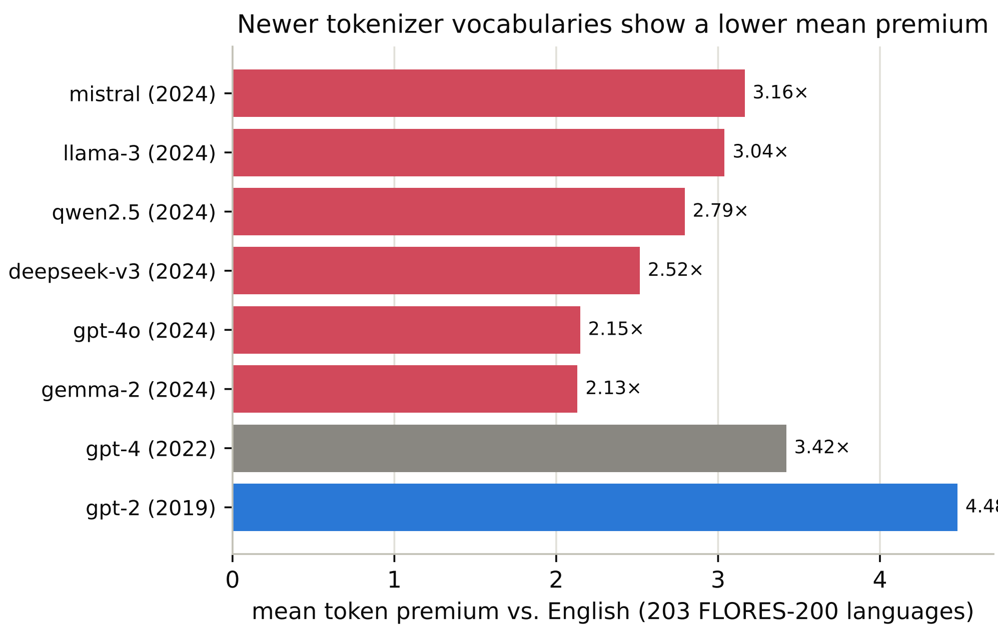

Figure 2: Mean token premium by tokenizer, colored by approximate release era of the associated model family (blue: 2019, gray: 2022, red: 2024). Release years reflect the model family's public release, not necessarily the exact date the tokenizer vocabulary itself was frozen.

5.4 Domain robustness

FLORES-200 is encyclopedic/news-register text; a premium ratio that only replicates on that one register is a weaker claim than one that holds across registers. For the 30 languages with the highest mean FLORES premium, we additionally measured the premium on a second corpus where one

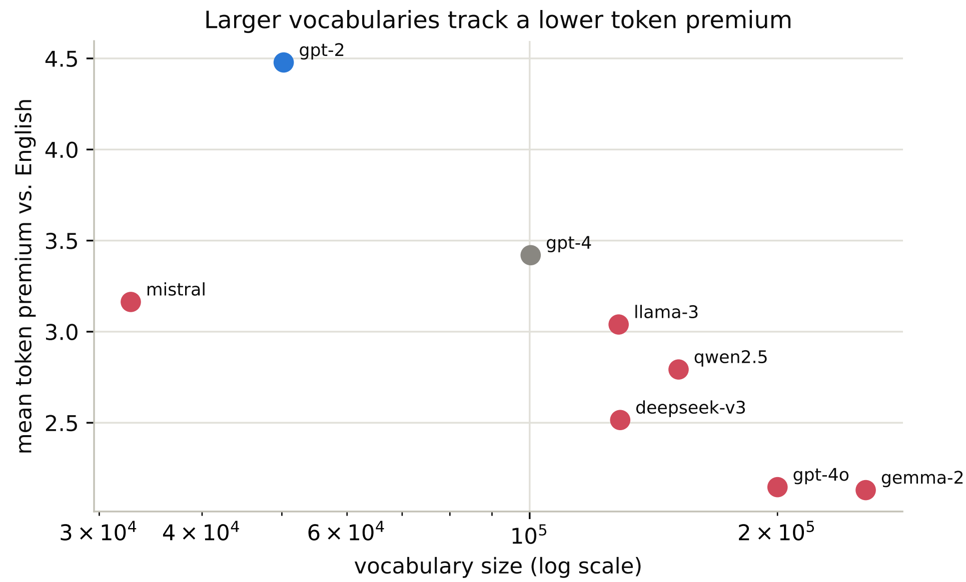

Figure 3: Mean token premium against vocabulary size (log scale), colored by release era as in Figure 2. Mistral (2024, 32,768 tokens) sits well above the 2024 tokenizers with larger vocabularies, and GPT-2 is both the oldest and among the smallest – vocabulary size tracks the premium more closely than release year does.

exists: mixed-domain OPUS-100 text and/or a Bible-translation corpus (religious register), each with its own aligned English pairing (not FLORES's).

Of these 30 languages, 21 have at least one second-domain corpus available; the other 9 – Shan, Santali, Tibetan, Tamasheq, Tamazight, Lao, Tigrinya, Manipuri, and Kabiyè – have no modern, ungated, actively-maintained parallel corpus on Hugging Face for a second domain at all. We report this gap explicitly rather than silently narrowing the check to the languages that happen to have data: the languages with the highest token tax are, by this evidence, also among those with the least available parallel data to independently corroborate it, which is itself a relevant fact for multilingual-NLP resource allocation.

Across the 264 (language, tokenizer, domain) comparisons where a second corpus exists, the median *absolute* relative difference from the FLORES estimate is 14% (mean 17%). Reporting only that figure would be misleading, because the two domains behave very differently once the sign is retained.

On OPUS-100 the difference is genuinely symmetric noise: median signed difference −1.7%, with 57% of comparisons falling below the FLORES estimate – about what chance would give. On the Bible corpus it is not noise at all. The median signed difference is −17.2%, and the domain premium is lower than the FLORES premium in *all 112* comparisons. A sign test against the null of symmetric divergence gives $p < 10^{-30}$; this is a systematic register effect, not sampling variation.

We therefore read the Bible-corpus result as a finding rather than a caveat: on religious-register parallel text, the token premium is consistently around a sixth lower than on FLORES's encyclopedic register. The plausible mechanism is that these translations use a smaller, more repetitive, more standardized vocabulary than news and encyclopedia prose, which favors any subword tokenizer in the non-English language; but we have not tested that, and it is equally consistent with these particular translations being older and more heavily edited than the FLORES sentences. The practical implication for users of this dataset is that a premium ratio is a property of a (tokenizer, language, register) triple, not of a (tokenizer, language) pair, and quoting one should say which register it came from.

Individual outliers remain worth flagging: Kannada on OPUS-100 diverges up to 107%, most likely a short, UI-string-heavy test split rather than a general property of the language, and Uyghur and Sanskrit on the Bible corpus diverge 44–72%, beyond even that domain's systematic shift.

5.5 Cost analysis

Following Ahia et al. [2023]'s framing, Table 3 converts premium ratios into dollar terms: at the $2.50-per-million-input-token figure for GPT-4o in our released pricing table – a placeholder snapshot explicitly marked unverified against current provider pricing, not a confirmed current price (see Limitations) – the same 1-million-token quantity of content costs $2.50 in English but up to $34.82 in Santali – a 13.9× markup for identical meaning, before any consideration of the additional latency and context-window cost of the extra tokens.

Language	Premium (GPT-4o)	Cost for content worth 1M English tokens
Santali	13.93×	$34.82
Tamasheq	10.41×	$26.04
Tamazight	10.26×	$25.65
Dzongkha	9.22×	$23.06
Lao	8.38×	$20.94
Tibetan	8.23×	$20.56
Shan	8.10×	$20.26
Tigrinya	6.05×	$15.13
Amharic	5.85×	$14.63
Odia	5.05×	$12.62
English (baseline)	1.00×	$2.50

Table 3: Cost to process 1 million English-tokens' worth of equivalent content, by language, at GPT-4o's input-token price in our released pricing table. Pricing is an unverified placeholder snapshot, not a confirmed current price; see Limitations.

6 Limitations

The premium we report is a mean of per-sentence ratios, which is upward-biased by roughly 1–2% relative to the ratio-of-totals estimand that prior work generally reports (Section 3). Comparisons against published figures should account for this; both quantities are recoverable from the released per-sentence data.

The vocabulary-size finding (Section 5.3) rests on 8 tokenizers, which is a small sample for a correlation, and vocabulary size is not randomly assigned – it covaries with how much multilingual data the vocabulary was trained on, which we cannot observe for six of the eight. Both pre-2024 baselines are also OpenAI vocabularies, partly confounding "older" with vendor in the temporal comparison. We report the association and the one case that discriminates between the two explanations (Mistral), not a controlled result.

The domain-robustness check (Section 5.4) covers 21 of the 30 highest-premium languages, not all 203, and the 9 uncovered languages are systematically the most extreme cases rather than a random sample – so neither the OPUS-100 agreement nor the Bible-corpus offset should be extrapolated to languages outside the 21 actually checked. All three corpora consist of translated rather than natively-authored text, which may introduce a shared translationese effect that a monolingual comparison would not have; the Bible-corpus offset we report is a difference between two translated registers, not between translated and natural text.

Pricing figures (Section 5.5) are a static, manually-maintained snapshot explicitly marked `verified: false` in the released pricing table; the library warns on load until a maintainer confirms current provider pricing, and dollar figures should be treated as illustrative of magnitude, not as a citable current price. Claude is excluded from every result in this paper for the reason given in Section 3, not because it is exempt from the phenomenon. The Llama 3 tokenizer is measured via an ungated third-party mirror, not Meta's own gated repository. Gemma-2 is reported only as itself, not as a validated proxy for Gemini's (unpublished) tokenizer. Finally, every number in this paper reflects specific tokenizer commits, recorded in the released code (Section 8); the five Hugging Face-hosted tokenizers were originally measured against their `main` branch rather than a frozen commit, which we have since corrected by pinning the registry to the exact commit `main` resolved to at that time (verified to reproduce identical token counts). Tokenizer vocabularies are periodically updated by their maintainers, and the versioned dataset, not this PDF, is the source of truth for current numbers.

7 Ethics Statement

This work is a measurement, not a system: it does not collect, generate, or redistribute personal data, and the released dataset contains only token counts, premium ratios, and confidence intervals – never the underlying FLORES-200, OPUS-100, or Bible-corpus sentence text, which remain under their original sources' own licenses. The motivating concern is fairness: speakers of the languages in Table 3 and Figure 1 pay materially more for equivalent access to commercial language models, a disparity this paper aims to make legible and citable rather than to newly create. We are not aware of a plausible dual-use or harm pathway specific to this artifact; the intended use is informing tokenizer design, pricing-policy discussion, and multilingual-NLP methods sections that need a current, citable number.

8 Availability

`tokentax` is released under the MIT license (results dataset under CC0) and available via `pip install tokentax` (`https://pypi.org/project/tokentax/`). Source, the full sweep and domain-robustness scripts, and tests are at `https://github.com/Shreyaskc/token-tax`. The versioned results dataset – 1.64M raw per-sentence rows, the CI-backed summary table, and the domain-robustness comparisons – is at `https://huggingface.co/datasets/shreyaskc/tokentax-results-v1`, and an interactive heatmap and cost calculator at `https://huggingface.co/spaces/shreyaskc/tokentax-explorer`. Continuous integration runs the test suite across Python 3.9–3.12 on every push.

9 Conclusion

The token tax is not a 2023 finding that has since been fixed. Across the seven currently-deployed tokenizers we measured, it persists at a mean of 2.74× and reaches 15× for the worst (language, tokenizer) pair. Even the two lowest-premium 2024 vocabularies still charge Georgian, Kannada, or

Malayalam speakers roughly 2–4× English's rate for identical content, and Qwen2.5 – released the same year – charges those same three languages 5–7×.

The more useful summary is not that tokenizers got better with time but that they got better with vocabulary size: across our set, vocabulary size predicts the premium substantially better than release year, and the 2024 tokenizer with the smallest vocabulary is barely an improvement on a vocabulary two years its senior. That reframing matters for anyone choosing a model for a non-English deployment, and it is actionable in a way that "newer is fairer" is not.

What will keep changing is the number itself. `tokentax` is built to be re-run, not re-derived: the measurement suite, the versioned dataset, and the explorer are designed so that the next tokenizer release gets a new, citable number within days, not another ad hoc re-implementation of Petrov et al.'s original analysis.

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